Appendix

Appendix 1: Examples of Search Strings by Language, Topic, and Database

Search strings in English:

1. Delays: 99 in total
    a. PubMed
        i. Search conducted on November 15, 2020
        ii. ((((“development* delay” OR “cognit* delay” OR “language delay” OR “ling* delay” OR “mot* delay” OR “emotion* delay”) AND (child* OR infant*)) AND “rural China”) AND “early childhood”)
        iii. Number of search results: 8
    b. EconPapers
        i. Search conducted on November 15, 2020
        ii. ((((“development* delay” OR “cognit* delay” OR “language delay” OR “ling* delay” OR “mot* delay” OR “emotion* delay”) AND (child* OR infant*)) AND “rural China”) AND “early childhood”)
        iii. Number of search results: 55
    c. PsycARTICLES
        i. Search conducted on November 15, 2020
        ii. ((((“development* delay” OR “cognit* delay” OR “language delay” OR “ling* delay” OR “mot* delay” OR “emotion* delay”) AND (child* OR infant*)) AND “rural China”) AND “early childhood”)
        iii. Number of search results: 5
    d. Cochrane Library
        i. Search conducted on November 15, 2020
        ii. TITLE-ABS-KEY(((((“development* delay” OR “cognit* delay” OR “language delay” OR “ling* delay” OR “mot* delay” OR “emotion* delay”) AND (child* OR infant*)) AND “rural China”) AND “early childhood”))
        iii. Number of search results: 1
    e. Web of Science
        i. Search conducted on November 15, 2020
        ii. TOPIC: ((((((“development* delay” OR “cognit* delay” OR “language delay” OR “ling* delay” OR “mot* delay” OR “emotion* delay”) AND (child* OR infant*)) AND “rural China”) AND “early childhood”)))
        iii. Number of search results: 17
    f. Scopus (Elsevier)
        i. Search conducted on November 15, 2020
        ii. TITLE-ABS-KEY(((((“development* delay” OR “cognit* delay” OR “language delay” OR “ling* delay” OR “mot* delay” OR “emotion* delay”) AND (child* OR infant*)) AND “rural China”) AND “early childhood”))
        iii. Number of search results: 13
2. Parenting: 110 in total
    a. PubMed
        i. Search conducted on November 15, 2020
        ii. ((((parenting OR caregiving) AND (child* OR infant*)) AND “rural China”) AND “early childhood”)
        iii. Number of search results: 30
    b. EconPapers
        i. Search conducted on November 15, 2020
        ii. ((((parenting OR caregiving) AND (child* OR infant*)) AND “rural China”) AND “early childhood”)
        iii. Number of search results: 25
    c. PsycARTICLES
        i. Search conducted on November 15, 2020

Emmers D, *et al. BMJ Global Health* 2021; 6:e005578. doi: 10.1136/bmjgh-2021-005578

ii. ((((parenting OR caregiving) AND (child* OR infant*)) AND “rural China”) AND “early childhood”)
iii. Number of search results: 25

d. Cochrane Library
i. Search conducted on November 15, 2020
ii. TITLE-ABS-KEY(((((parenting OR caregiving) AND (child* OR infant*)) AND “rural China”) AND “early childhood”))
iii. Number of search results: 4

e. Web of Science
i. Search conducted on November 15, 2020
ii. TOPIC: ((((((parenting OR caregiving) AND (child* OR infant*)) AND “rural China”) AND “early childhood”)))
iii. Number of search results: 13

f. Scopus (Elsevier)
i. Search conducted on November 15, 2020
ii. TITLE-ABS-KEY(((((parenting OR caregiving) AND (child* OR infant*)) AND “rural China”) AND “early childhood”))
iii. Number of search results: 13

3. Parenting interventions: 39 in total
a. PubMed
i. Search conducted on November 15, 2020
ii. (((((intervention* OR trial* OR RCT* OR experiment*) AND (child* OR infant*)) AND (stimulation OR nutrition)) AND “rural China”) AND “early childhood”)
iii. Number of search results: 14

b. EconPapers
i. Search conducted on November 15, 2020
ii. (((((intervention* OR trial* OR RCT* OR experiment*) AND (child* OR infant*)) AND (stimulation OR nutrition)) AND “rural China”) AND “early childhood”)
iii. Number of search results: 1

c. PsycARTICLES
i. Search conducted on November 15, 2020
ii. (((((intervention* OR trial* OR RCT* OR experiment*) AND (child* OR infant*)) AND (stimulation OR nutrition)) AND “rural China”) AND “early childhood”)
iii. Number of search results: 4

d. Cochrane Library
i. Search conducted on November 15, 2020
ii. (((((intervention* OR trial* OR RCT* OR experiment*) AND (child* OR infant*)) AND (stimulation OR nutrition)) AND “rural China”) AND “early childhood”)
iii. Number of search results: 3

e. Web of Science
i. Search conducted on November 15, 2020
ii. (((((intervention* OR trial* OR RCT* OR experiment*) AND (child* OR infant*)) AND (stimulation OR nutrition)) AND “rural China”) AND “early childhood”)
iii. Number of search results: 12

f. Scopus (Elsevier)
i. Search conducted on November 15, 2020
ii. (((((intervention* OR trial* OR RCT* OR experiment*) AND (child* OR infant*)) AND (stimulation OR nutrition)) AND “rural China”) AND “early childhood”)
iii. Number of search results: 5

Search strings in Chinese:

4. Delays: 257 in total
a. 中国知网 （CNKI）
i. 高级检索：学术期刊+ 学位论文 + 特色期刊
ii. 表达式

1. 主题：发育 + 认知 + 语言 + 运动 + 情感
2. AND 主题：迟缓 + 缓慢 + 滞后
3. AND 主题：婴儿 + 婴幼儿
4. AND 主题：农村

iii. 结果：共 73 篇

b. 万方数据 （Wanfang）
   i. 专业检索：期刊论文+ 学位论文
   ii. 来源数据库： 万方
   iii. 表达式：主题:(发育 + 认知 + 语言 + 运动 + 情感)*主题:(迟缓 + 缓慢 + 滞后 + 障碍)*主题:(婴儿 + 婴幼儿)*主题:(农村)
   iv. 结果：89 条

c. 维普（VIP）
   i. 检索式检索
   ii. 表达式：R=(发育 OR 认知 OR 语言 OR 运动 OR 情感) AND R=(迟缓 OR 缓慢 OR 滞后) AND R=(婴儿 OR 婴幼儿) AND R=(农村)
   iii. 结果：95 条

5. Parenting: 171 in total
   a. 中国知网 （CNKI）
      i. 高级检索：学术期刊 + 学位论文 + 特色期刊
      ii. 表达式
         1. 主题：养育 + 照顾 + 照料
         2. AND 主题：婴儿 + 婴幼儿
         3. AND 主题：农村

      iii. 结果：共 68 篇

   b. 万方数据 （Wanfang）
      i. 专业检索：期刊论文 + 学位论文
      ii. 来源数据库： 万方
      iii. 表达式：主题:(养育 + 照顾 + 照料)*主题:(婴儿 + 婴幼儿)*主题:(农村)
      iv. 结果：66 条

   c. 维普（VIP）
      i. 检索式检索
      ii. 表达式：R=(养育 OR 照顾 OR 照料) AND R=(婴儿 OR 婴幼儿) AND R=(农村)
      iii. 结果：37 条

6. Parenting interventions: 17 in total
   a. 中国知网 （CNKI）
      i. 高级检索：学术期刊 + 学位论文 + 特色期刊
      ii. 表达式
         1. 主题：干预 + 试验 + 随机对照试验 + 实验
         2. AND 主题：婴儿 + 婴幼儿
         3. AND 主题：养育 + 照顾 + 照料
         4. AND 主题：农村

      iii. 结果：6 条

   b. 万方数据 （Wanfang）
      i. 专业检索：期刊论文 + 学位论文
      ii. 来源数据库： 万方
      iii. 表达式：主题:(干预 + 试验 + 随机对照试验 + 实验)*主题:(婴儿 + 婴幼儿)*主题:(养育 + 照顾 + 照料)*主题:(农村)
      iv. 结果：11 条

   c. 维普（VIP）
      i. 检索式检索

Emmers D, *et al. BMJ Global Health* 2021; 6:e005578. doi: 10.1136/bmjgh-2021-005578

ii. 表达式：R=(干预 OR 试验 OR 随机对照试验 OR 实验) AND R=(婴儿 OR 婴幼儿) AND R=(养育 OR 照顾 OR 照料) AND R=(农村)
iii. 结果：0 条

Appendix 2: Prevalence of Developmental Delay

1. Data Extraction and Processing

Prevalence was extracted and processed in three ways:

1. If the reviewed study reported the prevalence of developmental delay, defined as a developmental score of more than 1 SD below the healthy population mean, we extracted the percentage of delay reported in the study; or

2. If the reviewed study reported the percentage of delay, but this percentage was defined based on a different cutoff score, we imposed the assumptions of a normal distribution to derive the percentage of children with a developmental score of more than 1 SD below the healthy population mean. For example, if the study defined developmental delay as a score of at least 2 SD below the mean of a healthy populations, we derived the percentage of children with a score of at least 1 SD below this mean, assuming that the developmental scores of the sample population follow a normal distribution; or

3. If the mean and SD of development scores were reported instead of the percentage of delay, we calculated the percentage of delay based on published norms of a representative reference population in combination with the assumption that the developmental scores of the sample population follow a normal distribution.

2. Measurement of Prevalence of Developmental Delay Based on BSID-III Scores

2.1. Normative sample

The ASQ-3, ASQ:SE, BSID-I, and BSID-II have been validated and standardized based on an urban Chinese normative sample.[25–29] No normative Chinese sample for the BSID-III scores, however, has been established. Therefore, the raw score conversion charts included in the original US BSID-III administration manual for each age category are used to compute BSID-III composite scores in the reviewed studies.[30]

Hua et al.[31] assessed BSID-III (composite and raw) cognition scores for over 1,500 urban Chinese children. As a robustness check, we can alternatively use the distribution of this urban Chinese sample to norm the BSID-III cognition scores. Hua et al. report the mean (SD) of BSID-III cognition scores for seven separate age groups. Disregarding the limited number of age groups, which limits the accuracy with which we can compute China-specific age-adjusted normed scores, Appendix Table 2.1 shows that using China-specific norms to define cognitive delay does not lead to a systematic increase or decline in reported prevalences of delay. Note that, in absence of a study that establishes the mean (SD) of BSID-III language and motor composite scores for a large-scale urban Chinese sample, we are unable to conduct the same robustness check for BSID-III language and motor delays.

2.2. Cutoff scores used to define developmental delay

Note that the BSID was designed to have a mean composite score of 100 with a SD of 15 for a healthy population.[30] Empirical evidence from population-based random samples globally finds higher means and lower SDs for the cognition, language, and motor scales of the third edition of the BSID (BSID-III).[25,32,33] The literature shows that, when using the traditional cutoff score of 85, the prevalence of moderate developmental delay is underestimated.[32,34] In absence of a normative Chinese sample, BSID-III cutoff scores are defined based on the means (SDs) of BSID-III composite scores reported in the international literature (i.e., mean (SD) is expected to be 105 (9.6), 109 (12.3), and 100 (15) for the cognitive, language, and social-emotional scales, respectively.[30,32,35,36] Finally, note that the recent study of Hua et al.[31] reported a mean (SD) of 102 (6.9) for BSID-III composite cognition scores of urban Chinese children. When we use this cutoff score instead of the cutoff score extracted from the global literature, the cutoff score remains nearly identical to cutoff scores reported in the reviewed international literature (i.e., 95.4 vs. 95.1).

Appendix Table 2.1: Prevalence of Cognitive Delay based on US vs. China BSID-III Norms

No.	Author(s) (Year)	Child ages (months)	Mean (SD) of BSID-III composite cognition scores (based on US norms)	Mean (SD) of BSID-III raw cognition scores	Prevalence of cognitive delay	
					US norms[a]	China norms[b]
1	Luo et al. (2019a)[8]	6–18	95.7 (12.9)	38.6 (8.6)	57%	66%
2	Emmers et al. (2020)[9]	6–18	100.5 (11.9)	43.5 (7.7)	41%	43%
3	Wang et al. (2020a)[10]	6–24	95.9 (12.6)	45.5 (11.2)	54%	38%
4	Qian et al. (2020)[11]	6–18	96.7 (13.4)	40.9 (10.3)	50%	55%
5	Wang et al. (2019)[14]	6–30	96.5 (13.3)	46.2 (11.6)	47%	36%
6	Zhong et al. (2019)[15]	1–23	94.5 (16.2)	43.6 (11.2)	56%	45%
7	Wang et al. (2020b)[18]	6–24	96.8 (13.1)	45.9 (11.2)	52%	37%
8	Emmers & Wang (2020)[19]	6–36	101.4 (14.9)	55.6 (14.2)	39%	47%

Note. a. US norms are defined based on the normative US sample.[30] If mean (SD) of the BSID-III raw cognition scores was not reported in the paper, we derived the raw scores using the conversion charts included in the original BSID-III administration manual for each age category.[30] If the sample covered multiple age categories, the median sample age was used. b. China norms are defined based on the distribution of Bayley-III raw cognition scores reported in Hua et al.[31] by age category. If the sample covered multiple age categories, the median sample age was used.

Appendix 3: Risk of Bias Assessments

Appendix Table 3.1: Risk of Bias Assessment for Prevalence Studies (Hoy et al., 2012)

Study details		External validity* (No = high risk; Yes = low risk)					Internal validity* (No = high risk; Yes = low risk)					Overall
Author(s)	**Year**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11 (Overall risk of bias)**
Jin et al.	2007[1]	No	Yes	Yes	No	Yes	No	Yes	Yes	Yes	No	Moderate
Ma et al.	2008[2]	Yes	Yes	No	No	Yes	Yes	Yes	Yes	No	Yes	Low
Yang et al.	2019[3]	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes	No	Low
Yue et al.	2017[4]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	No	Low
Zhou et al.	2019a[5]	Yes	Yes	No	No	Yes	No	Yes	Yes	Yes	Yes	Low
Zhang et al.	2018[6]	Yes	Yes	No	No	Yes	No	Yes	Yes	Yes	Yes	Low
Wei et al.	2018[7]	Yes	Yes	No	No	Yes	No	Yes	Yes	Yes	Yes	Low
Luo et al.	2019a[8]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Low
Emmers et al.	2020[9]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Low
Wang et al.	2020a[10]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Low
Qian et al.	2020[11]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	No	Low
Li et al.	2018[12]	Yes	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Low
Zhou et al.	2019b[13]	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Low
Wang et al.	2019[14]	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Low
Zhong et al.	2019[15]	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes	No	Low
Bai et al.	2020[16]	Yes	No	Yes	No	Yes	Yes	Yes	Yes	Yes	No	Low
Tan et al.	2020[17]	Yes	Yes	No	No	Yes	No	Yes	Yes	Yes	Yes	Low
Wang et al.	2020b[18]	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes	No	Low
Emmers & Wang	2020[19]	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	No	Low

Note. *Items of Hoy et al.'s (2012) risk of bias tool:
1. Was the study's target population a close representation of the rural population in relation to relevant variables, e.g., age, sex, occupation?
2. Was the sampling frame a true or close representation of the target population?
3. Was some form of random selection used to select the sample, OR was a census undertaken?
4. Was the likelihood of non-response bias minimal?
5. Were data collected directly from the subjects (as opposed to a proxy)?
6. Was an acceptable case definition used in the study?
7. Was the study instrument that measured the parameter of interest (e.g., prevalence of low back pain) shown to have reliability and validity (if necessary)?
8. Was the same mode of data collection used for all subjects?
9. Was the length of the shortest prevalence period for the parameter of interest appropriate?
10. Were the numerator(s) and denominator(s) for the parameter of interest appropriate?

Appendix Table 3.2: Risk of Bias Assessment of Intervention Studies (using EPHPP tool)

Author(s)	Year	A. Selection bias rating	B. Study design rating	C. Confounders rating	D. Blinding rating	E. Data collection methods rating	F. Withdrawals and dropout rating	Global rating
Jin et al.	2007[1]	3	3	3	1	3	3	3
Sylvia et al.	2020[20]	3	3	3	2	3	3	3
Wang et al.	2020c[21]	3	3	3	2	3	3	3
Heckman et al.	2020[22]	3	3	3	2	3	3	3
Luo et al.	2019b[23]	3	3	3	3	3	3	3
Emmers et al.	2020[9]	3	3	3	3	3	2	3
Zhong et al.	2020[24]	3	3	3	2	3	2	3
Qian et al.	2020[11]	3	3	3	2	3	2	3
Li et al.	2018[12]	2	3	3	2	3	1	2
Bai et al.	2020[16]	3	3	3	3	3	3	3

Note. The Effective Public Health Practice Project (EPHPP) quality assessment tool for quantitative studies is a standardized tool to assess the quality of intervention studies. The tool evaluates study quality based on the domains of selection bias, study design, confounders, blinding, data collection methods, withdrawals, and dropouts. The tool can be used to evaluate a variety of intervention studies, including RCTs and case-control studies. According to the grading guidelines, the study obtains 1, 2, and 3 points, respectively, if the study is rates as weak, moderate, or strong for a specific domain. Domain scores are averaged to compute the global rating score. The maximum global rating score is 3. Based on the global score, studies are assigned a global quality rating of weak (1–1.50), moderate (1.51–2.50), or strong (2.51–3).

Appendix 4: GRADE Summary of Findings

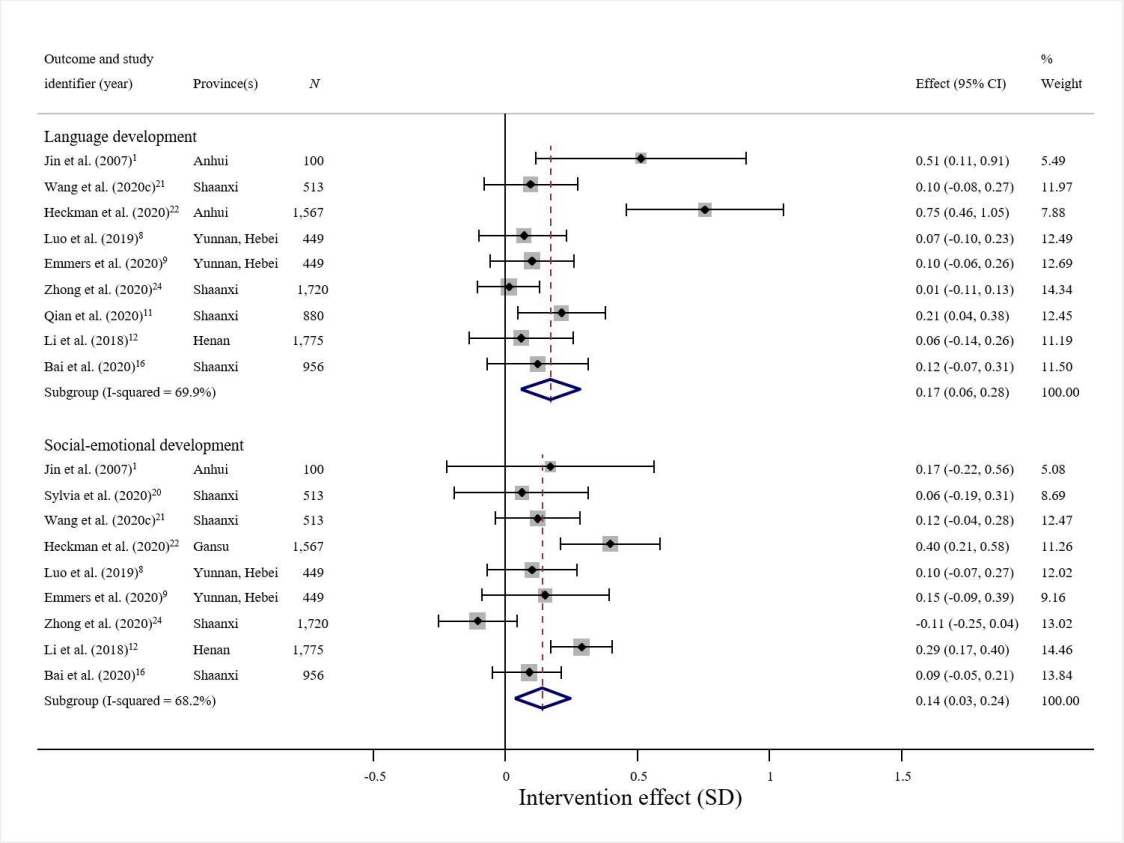
Appendix Table 4.1: GRADE Summary of Findings on Effects of Parental Training Programs on ECD Outcomes and Parenting Practices and Beliefs

	Certainty assessment							Summary of findings		
	Participants (no. of studies)	Study design	Risk of bias	Inc.	Ind.	Imp.	Pub. bias	Impact	Indicative effect sizes[a], SMD (95% CI)	Certainty
Primary outcomes										
Cognitive development	7,960 (10)	RCT	No	S(-1)	No	No	No	Positive	0.26 (0.18 to 0.35)	We have high certainty that parental training programs improve early cognition outcomes of 0- to 5-year-olds in rural areas of China[b]
Language development	7,960 (9)	RCT	No	S(-1)	No	No	No	Positive	0.17 (0.06 to 0.28)	We have moderate certainty that parental training programs improve early language outcomes of 0- to 5-year-olds in rural areas of China[c]
Social-emotional development	7,080 (9)	RCT	No	S(-1)	No	No	No	Positive	0.14 (0.03 to 0.24)	We have moderate certainty that parental training programs improve early social-emotional outcomes of 0- to 5-year-olds in rural areas of China[*d]
Secondary outcomes										
Parenting practices	4,518 (7)	RCT	No	S(-1)	S(-1)	No	No	Positive	0.39 (0.24 to 0.54)	We have moderate certainty that parental training programs improve early cognition outcomes of 0- to 5-year-olds in rural areas of China[e]
Parenting knowledge and beliefs	3,562 (6)	RCT	No	S(-1)	S(-1)	No	No	Positive	0.20 (0.11 to 0.28)	We have moderate certainty that parental training programs improve early cognition outcomes of 0- to 5-year-olds in rural areas of China[f]

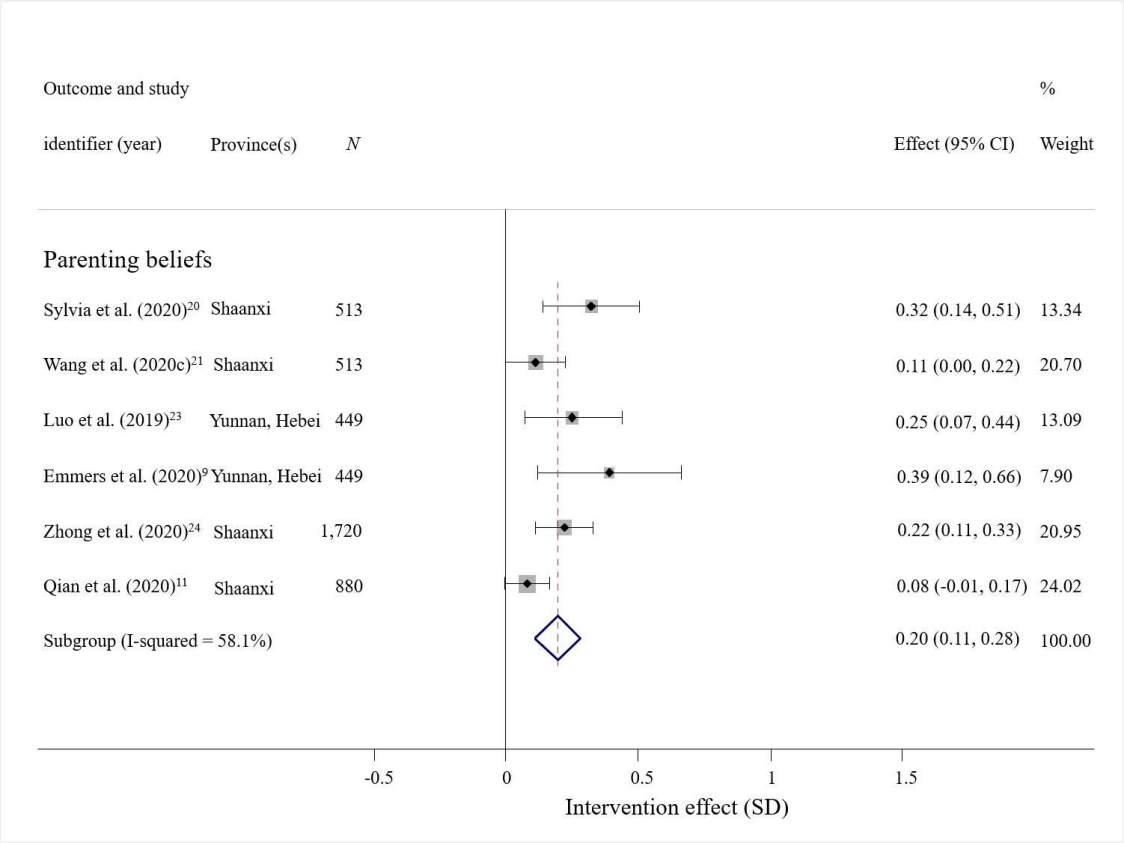
Note. GRADE: Grading of Recommendations Assessment, Development, and Evaluation; ECD: Early Childhood Development; SMD: standardized mean difference; S(-1): serious risk, downgrade by one point. Inc.: Inconsistency; Ind.: Indirectness; Imp,: Imprecision.
a. From meta-analysis; b. Rating down for inconsistency in outcome measures of parenting practices, up for dose-response estimates; c. Rating down for inconsistency in outcome measures of parenting practices; d. Rating down for inconsistency in outcome measures of parenting practices; e. Rating down for inconsistency and indirectness in outcome measures of parenting practices, up for dose-response estimation; f. Rating down for inconsistency and indirectness in outcome measures of parenting beliefs, up for dose-response estimation.

Appendix 5: Intervention Effects on Outcomes of Child Development and Parental Beliefs

Appendix Figure 5.1: Intervention Effects on Early Language and Social-Emotional Development



Appendix Figure 5.2: Intervention Effects on Parental Beliefs



Appendix 6: Heterogeneity in Intervention Effects

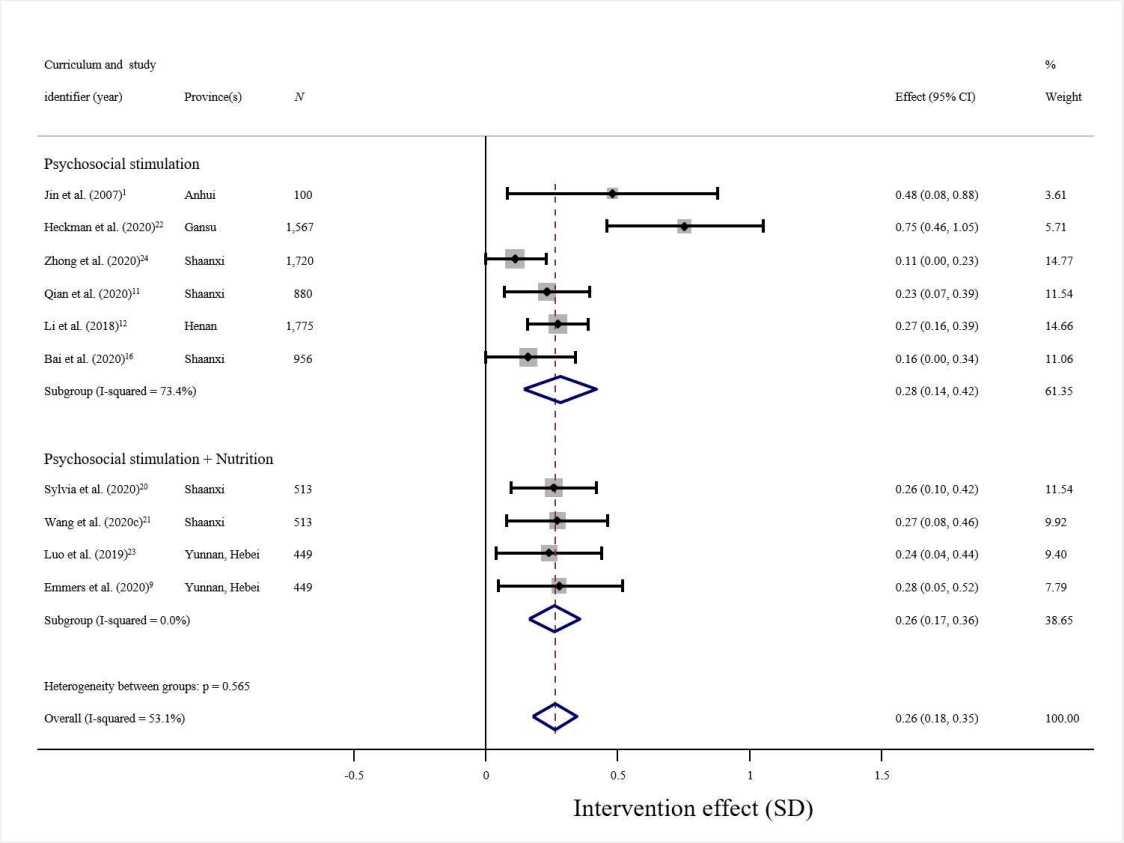
We conduct a heterogeneity analysis of treatment impacts by program curricula and delivery mode, measurement tool, and geographic location. Figures 6.1–6.18 in this Appendix document show the results of the heterogeneity testing. Figures 6.1–6.3 and Figures 6.4–6.6 present the heterogeneity in treatment impacts across studies with different program curricula and delivery modes, respectively. First, we find no evidence of systematic differences in treatment impacts across single-component psychosocial stimulation programs and integrated psychosocial stimulation and nutrition programs (*p*-values of between-group heterogeneity > 0.10; see Figures 6.1–6.3). Second, Figures 6.4–6.6 show that center-based programs have, on average, an impact of 0.19 SD [95% CI: 0.11–0.27 SD], 0.09 SD [95% CI: -0.005–0.18 SD], and 0.09 SD [95% CI: -0.13–0.32 SD on early cognitive, language, and social-emotional development, respectively. In contrast, home-based programs have, on average, an impact of 0.34 SD [95% CI: 0.21–0.48 SD], 0.27 SD [95% CI: 0.05–0.48 SD], and 0.17 SD [95% CI: 0.06–0.28 SD on child cognitive, language, and social-emotional development, respectively. We observe that treatment impact heterogeneity between center-based and home-based programs is significant at the 5 percent level for cognitive development but not for language or social-emotional development.

Figures 6.7–6.12 present heterogeneity in the prevalence of delay and the size of estimated treatment impacts on child development outcomes by type of ECD measurement tool. The results of our measurement-based heterogeneity analysis indicate that the point estimates of the prevalence of cognition and language delay are lower in studies in which the ASQ-3 test is used than in studies in which the BSID is administered (37.0% [95% CI: 24.0–50.1%] vs. 48.2% [95% CI: 45.3–51.1%] and 35.8% [95% CI: 25.2–46.3%] vs. 52.1% [95% CI: 47.1–58.3%]; see Figures 6.7 and 6.8 in Appendix 6, respectively). We do not detect a meaningful difference in the observed prevalence of social-emotional delay when using ASQ-3 tests, ASQ:SE tests, or the BSID (40.5% [95% CI: 29.7–51.3%] vs. 32.3% [95% CI: 23.1–41.4%] vs. 36.1% [95% CI: 27.4–44.8%], see Figure 6.9). Further, Figures 6.10–6.12 show that the parenting interventions have, on average, an impact of 0.20 SD [95% C:I 0.12–0.27 SD], 0.09 SD [95% CI: 0.00–0.17 SD], and 0.03 SD [95% CI: -0.09–0.16 SD] on BSID cognition, language, and social emotional scores, respectively. Note that only one of the reviewed impact evaluations used ASQ-3 scores to measure ECD outcomes.

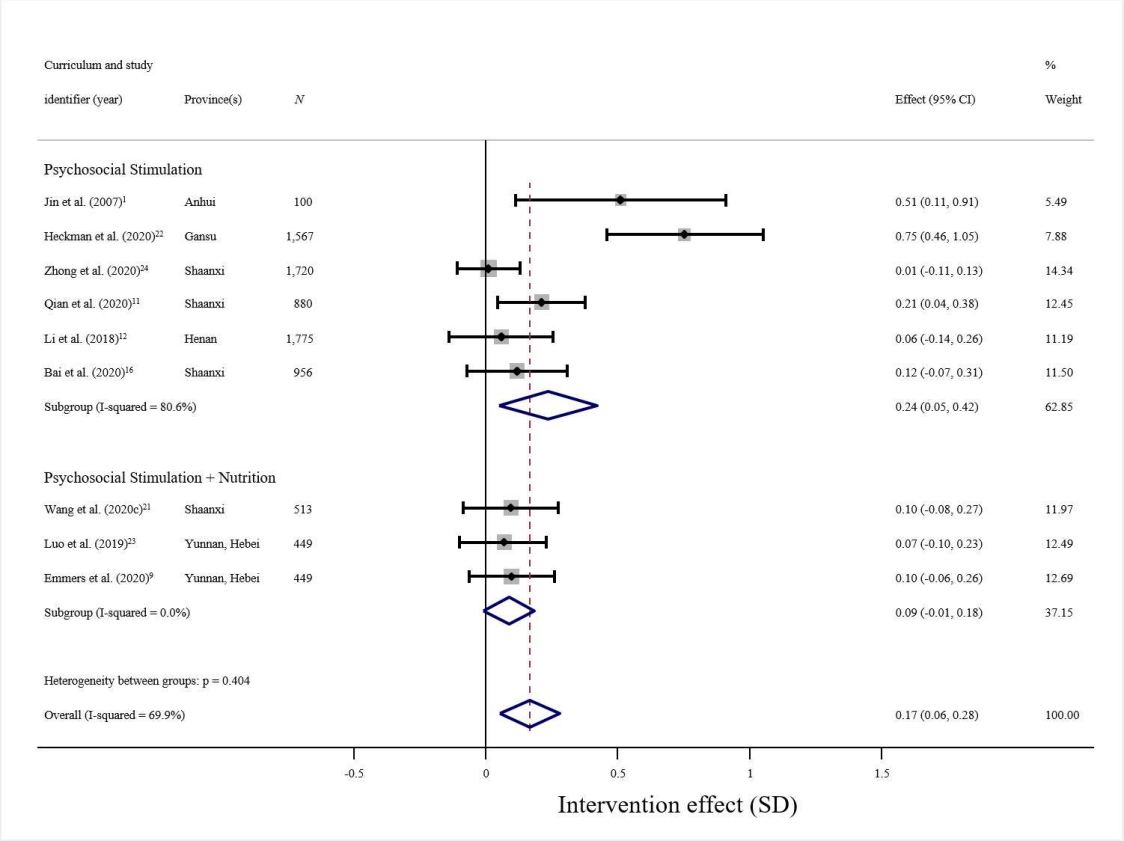
Finally, Figures 6.13–6.18 show the heterogeneity in the prevalence of delay and the size of estimated treatment impacts on child development outcomes by geographic location of the study. The reviewed studies span 14 provinces across Eastern, Central, and Western China. By grouping studies by geographic location, we find that the risk of early cognitive and language delay is higher in Western China than in Eastern and Central China (see Figures 6.13 and 6.14), but treatment impacts on child cognitive and language development do not differ significantly across these regions (*p*-values of between-group heterogeneity > 0.10; see Figures 6.16 and 6.17). Evidence from two studies conducted in Eastern China indicates that the prevalence of social-emotional delay and treatment impacts on social-emotional child outcomes in Eastern China are lower than in Central and Western China (see Figures 6.15 and 6.18). The evidence in Figures 6.15 and 6.18 suggests that the risk of social-emotional delay and the potential role for parenting interventions may be smaller in Eastern China. We acknowledge, however, that we are unable to draw conclusions with regard to geographic variation in our results due to data limitations of this study.

1. Heterogeneity by Program Curriculum

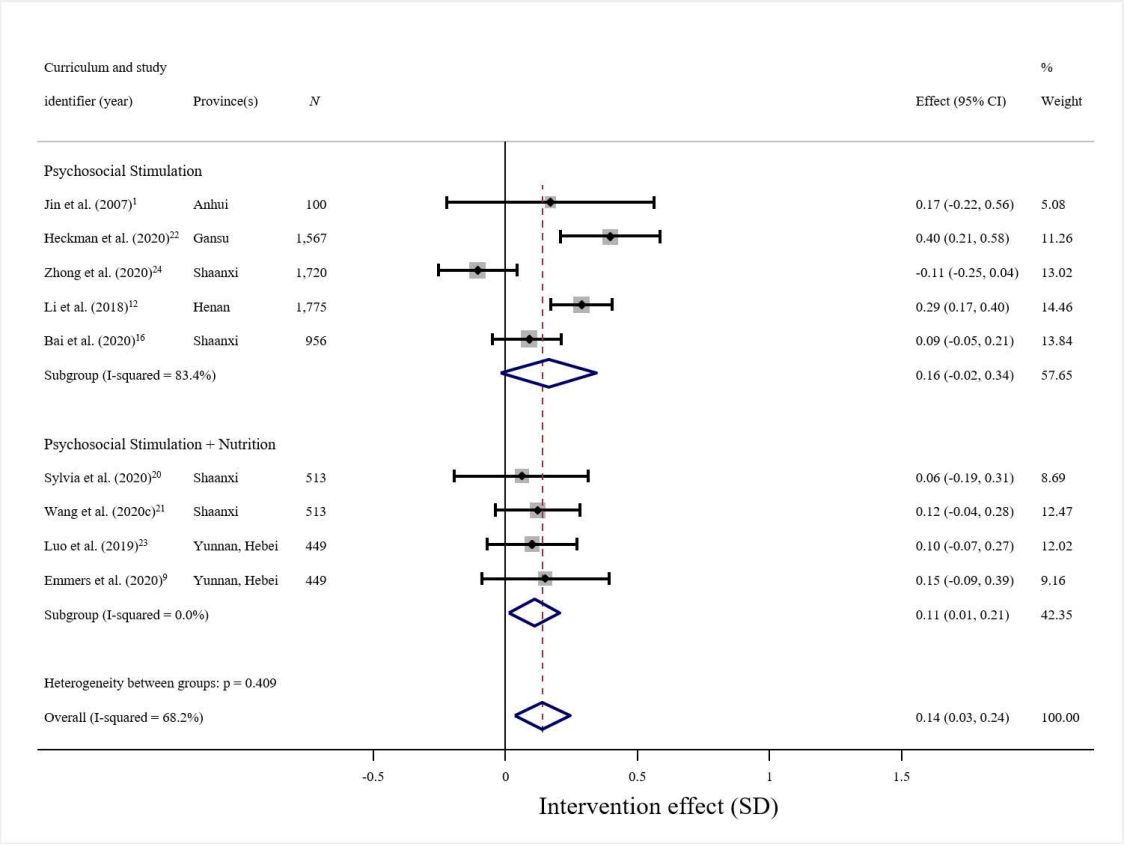
Appendix Figure 6.1: Pooled Intervention Effects on Early Cognitive Development by Program Curriculum



Appendix Figure 6.2: Pooled Intervention Effects on Early Language Development by Program Curriculum

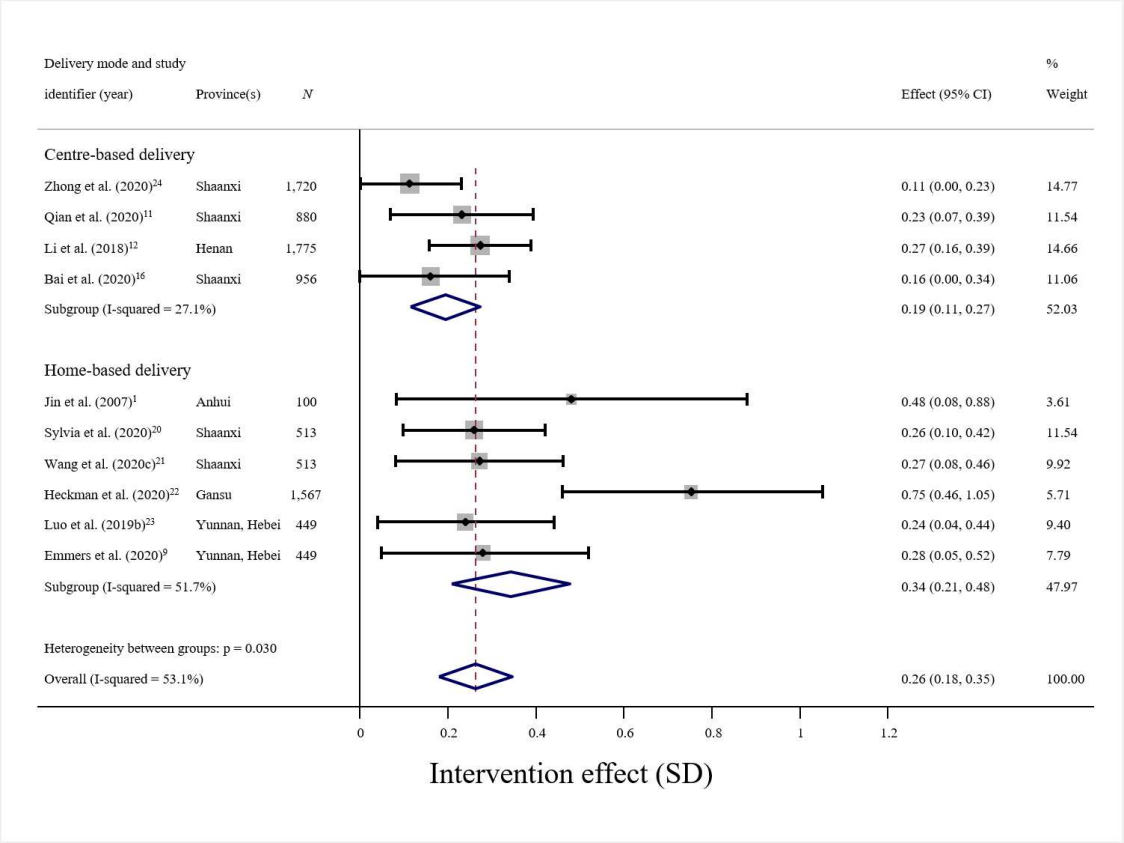


Appendix Figure 6.3: Pooled Intervention Effects on Early Social-Emotional Development by Program Curriculum

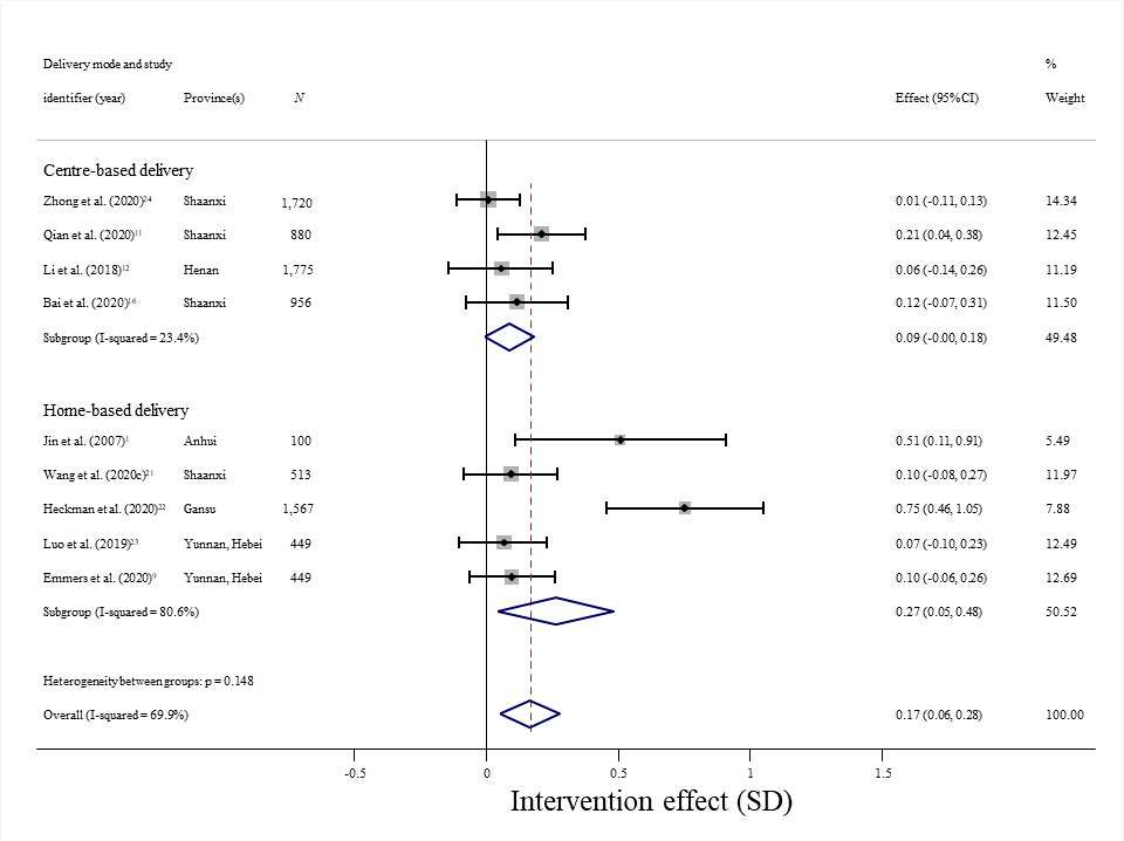


2. Heterogeneity by Delivery Mode

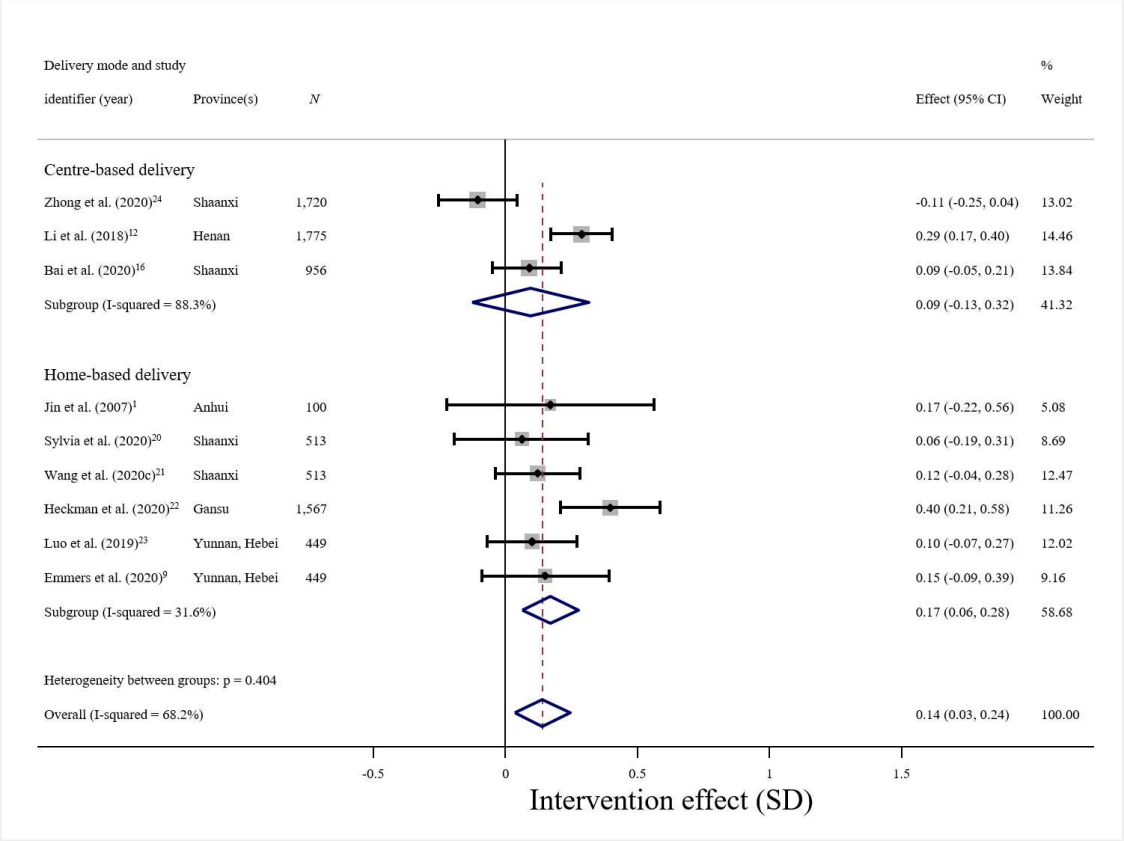
Appendix Figure 6.4: Pooled Intervention Effects on Early Cognitive Development by Delivery Mode



Appendix Figure 6.5: Pooled Intervention Effects on Early Language Development by Delivery Mode

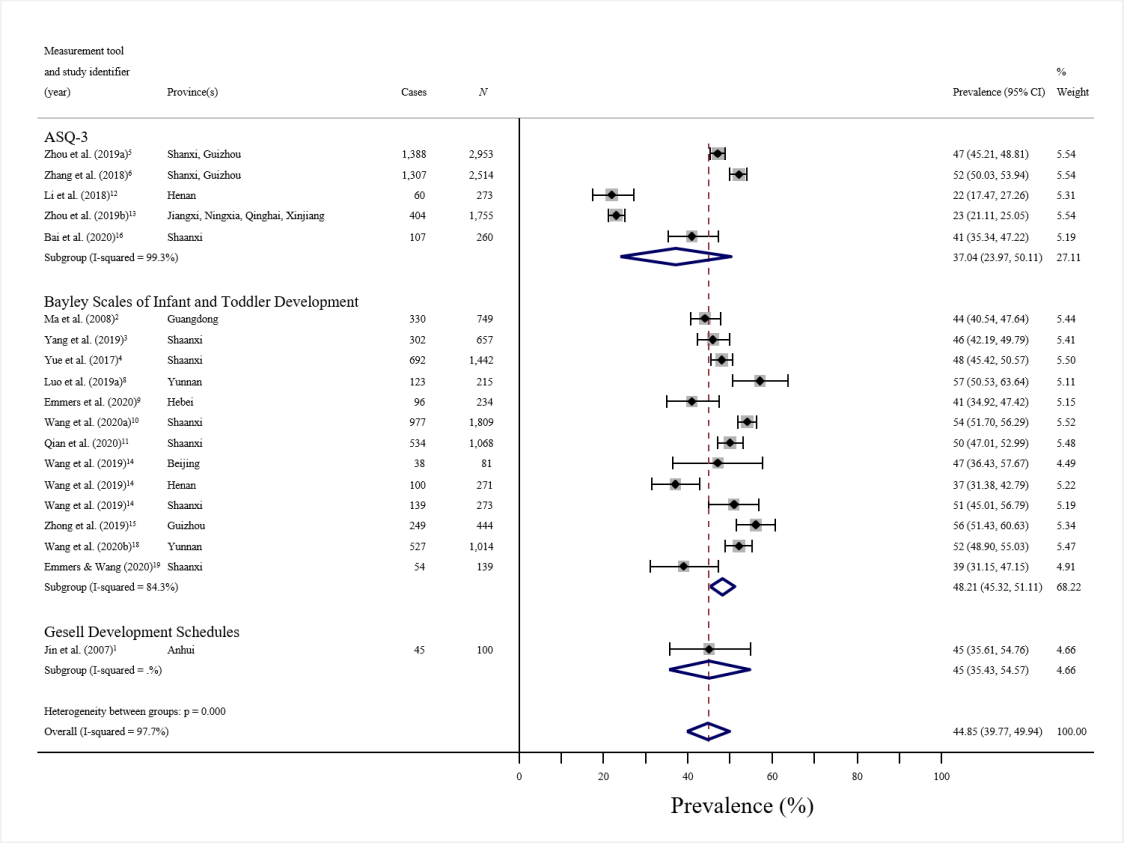


Appendix Figure 6.6: Pooled Intervention Effects on Early Social-Emotional Development by Delivery Mode

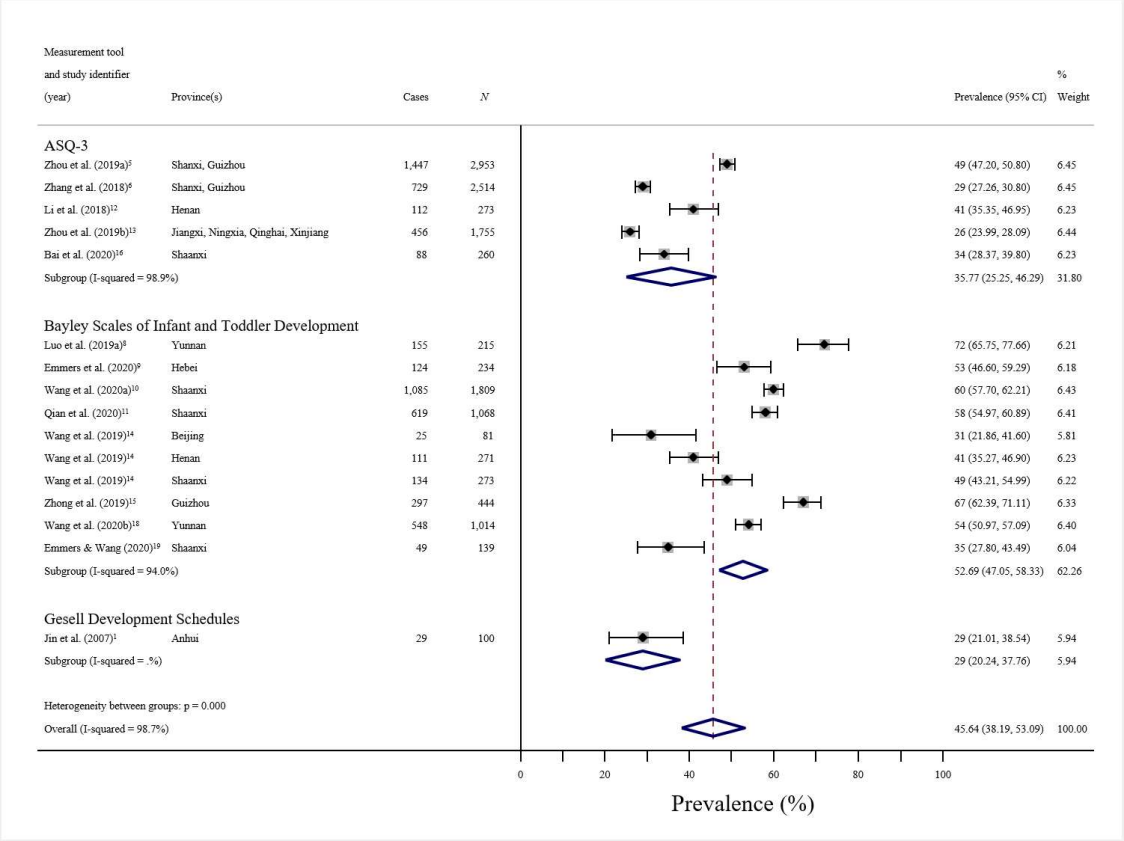


3. Heterogeneity by Measurement Tool

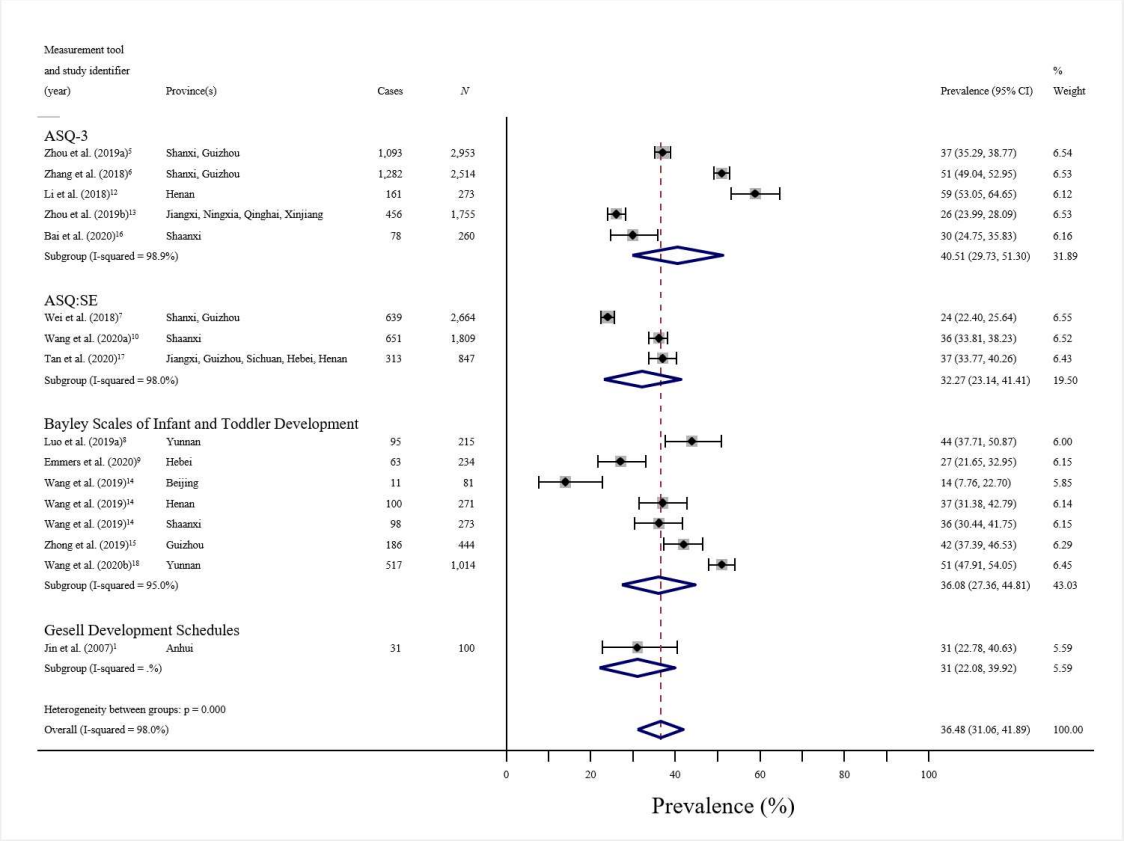
Appendix Figure 6.7: Pooled Prevalence of Early Cognitive Delay by Measurement Tool



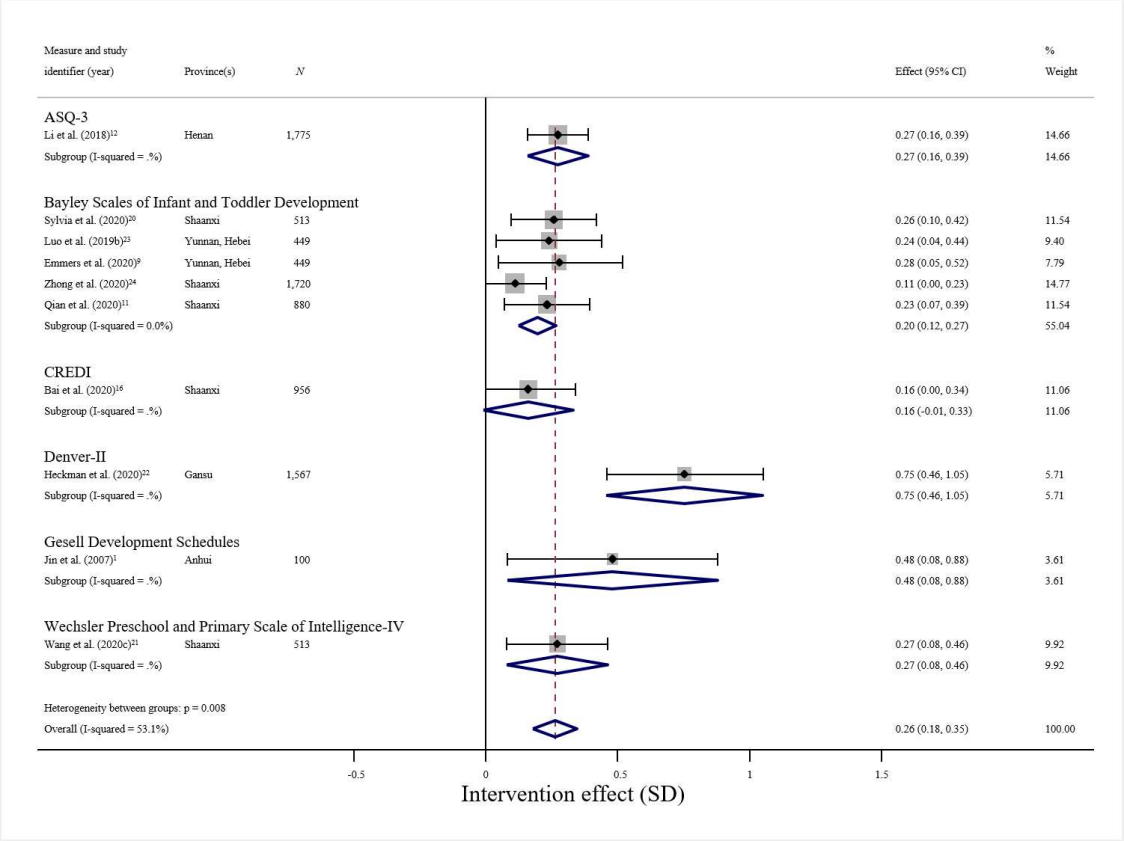
Appendix Figure 6.8: Pooled Prevalence of Early Language Delay by Measurement Tool



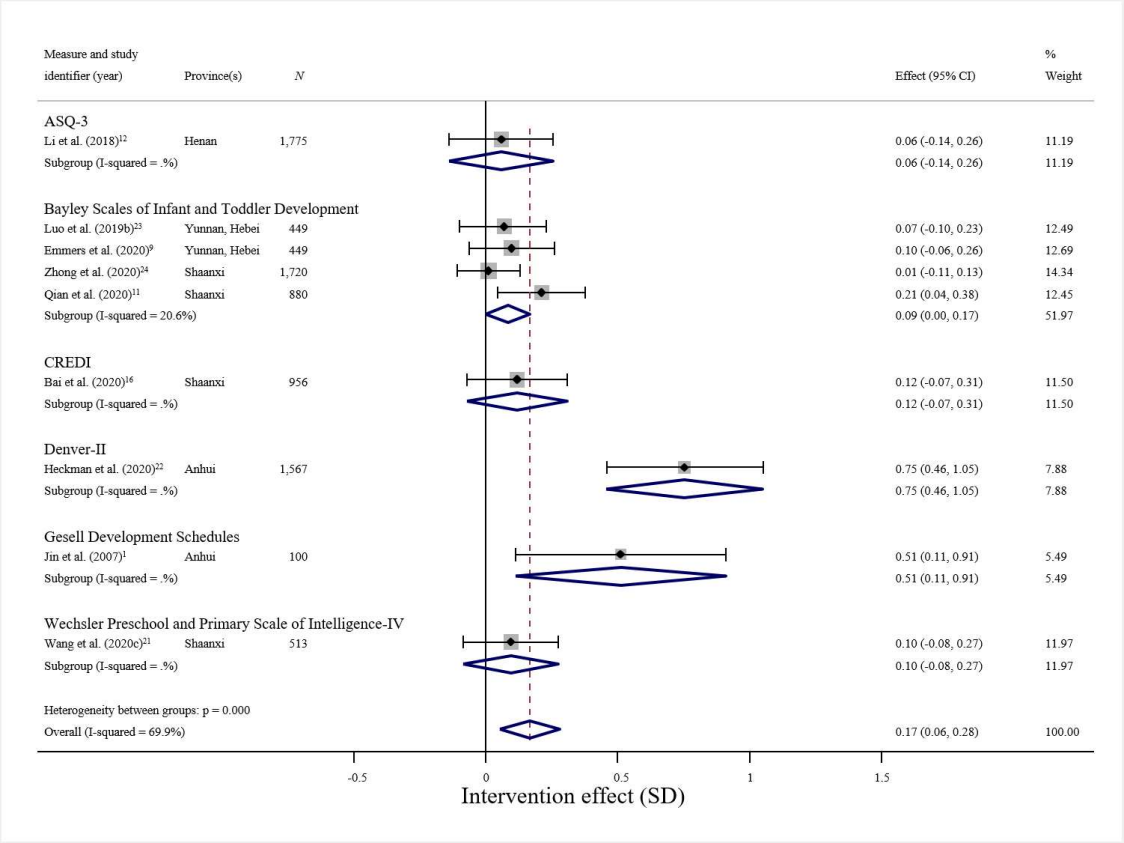
Appendix Figure 6.9: Pooled Prevalence of Early Social-Emotional Delay by Measurement Tool



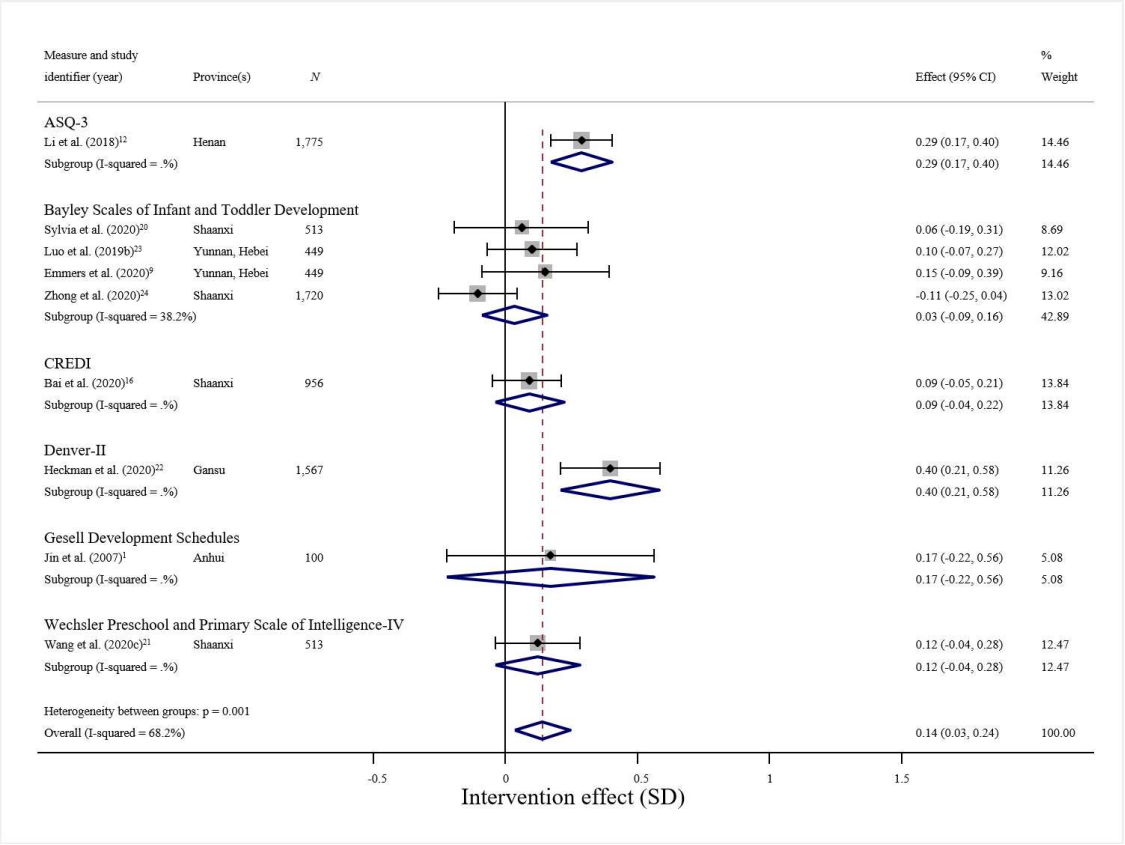
Appendix Figure 6.10: Pooled Intervention Effects on Early Cognitive Development by Measurement Tool



Appendix Figure 6.11: Pooled Intervention Effects on Early Language Development by Measurement Tool

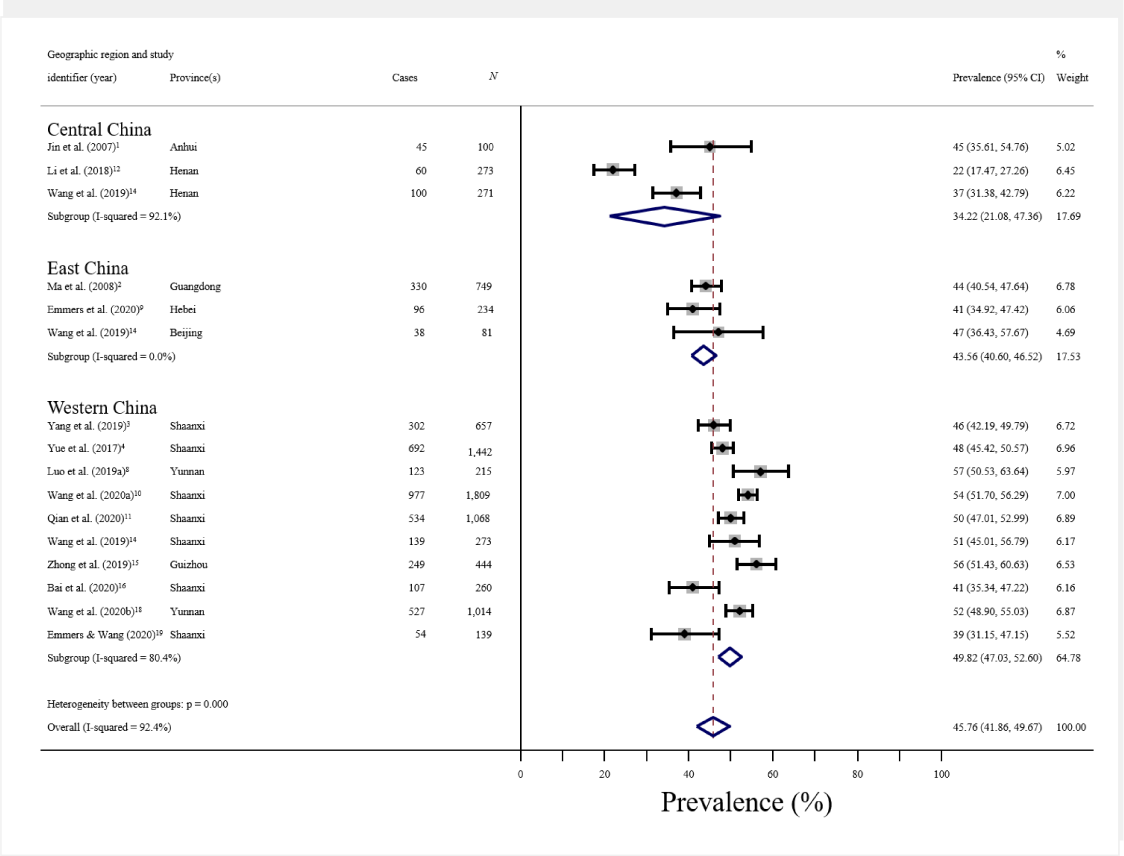


Appendix Figure 6.12: Pooled Intervention Effects on Early Social-Emotional Development by Measurement Tool

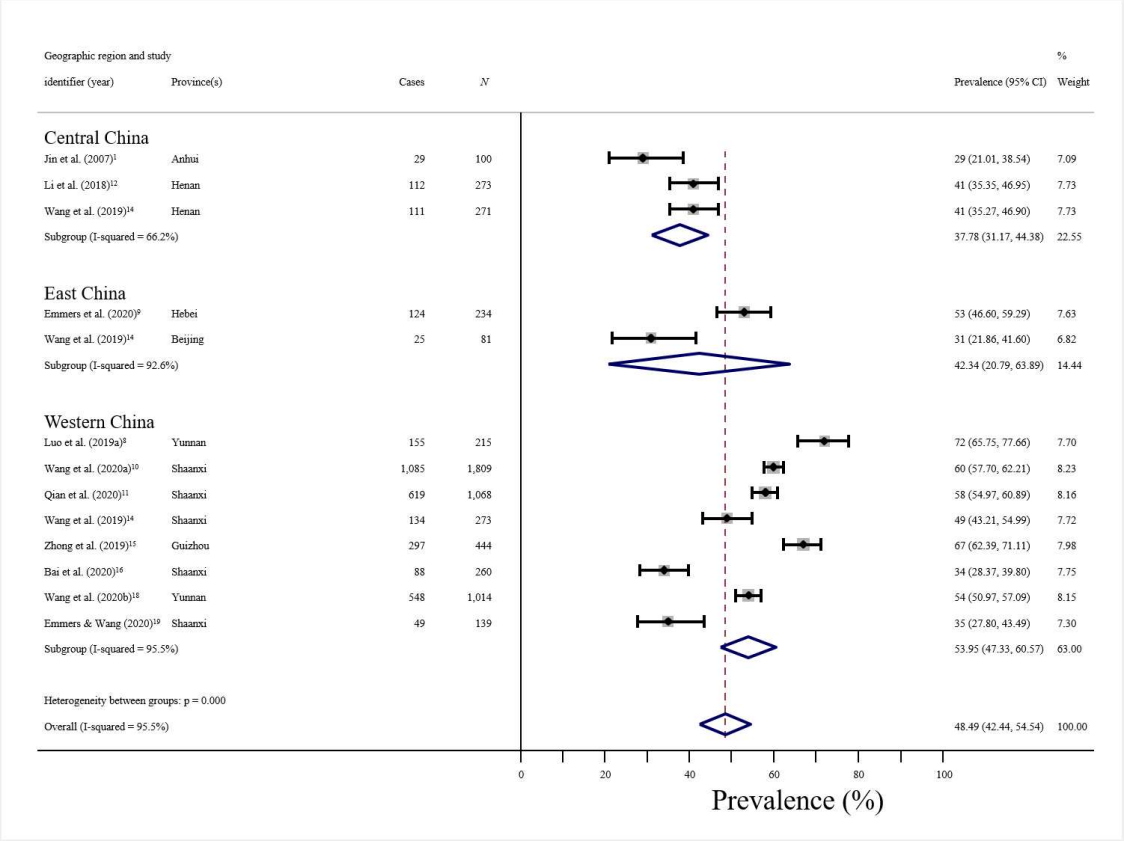


4. Heterogeneity by Geographic Region

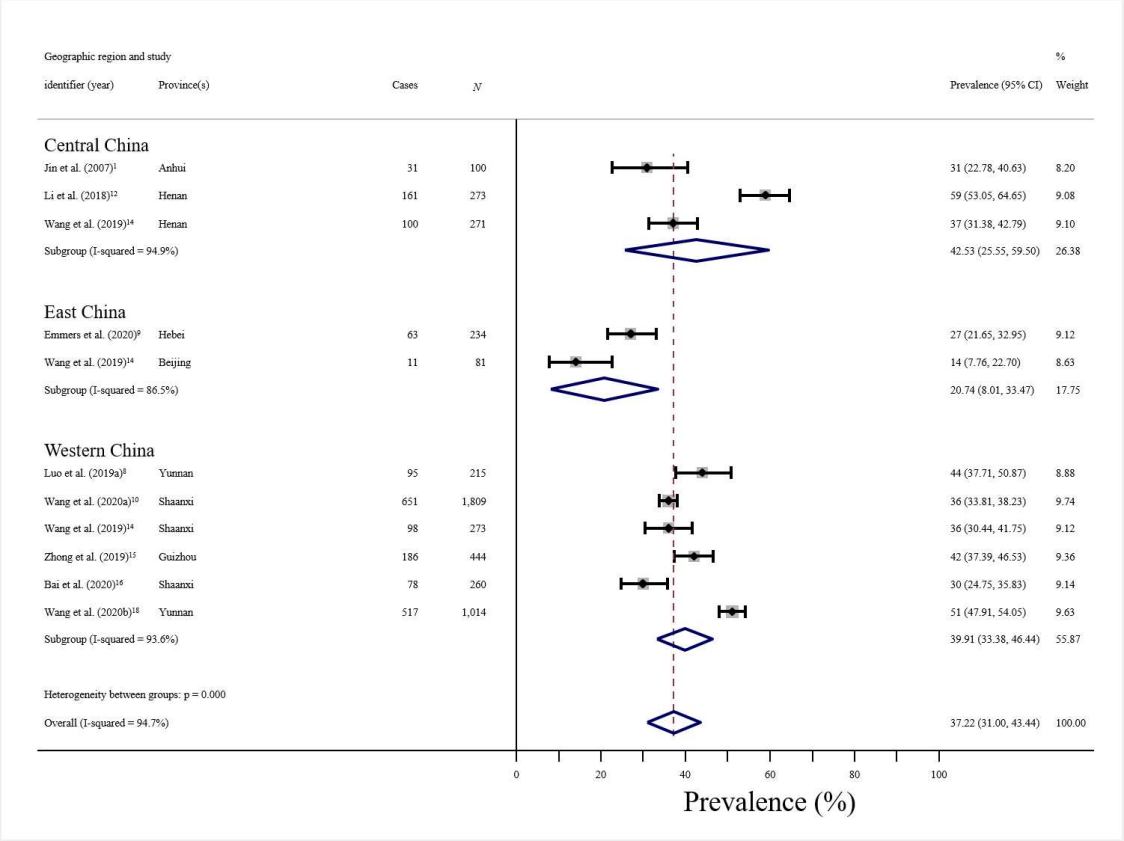
Appendix Figure 6.13: Pooled Prevalence of Early Cognitive Delay by Geographic Region



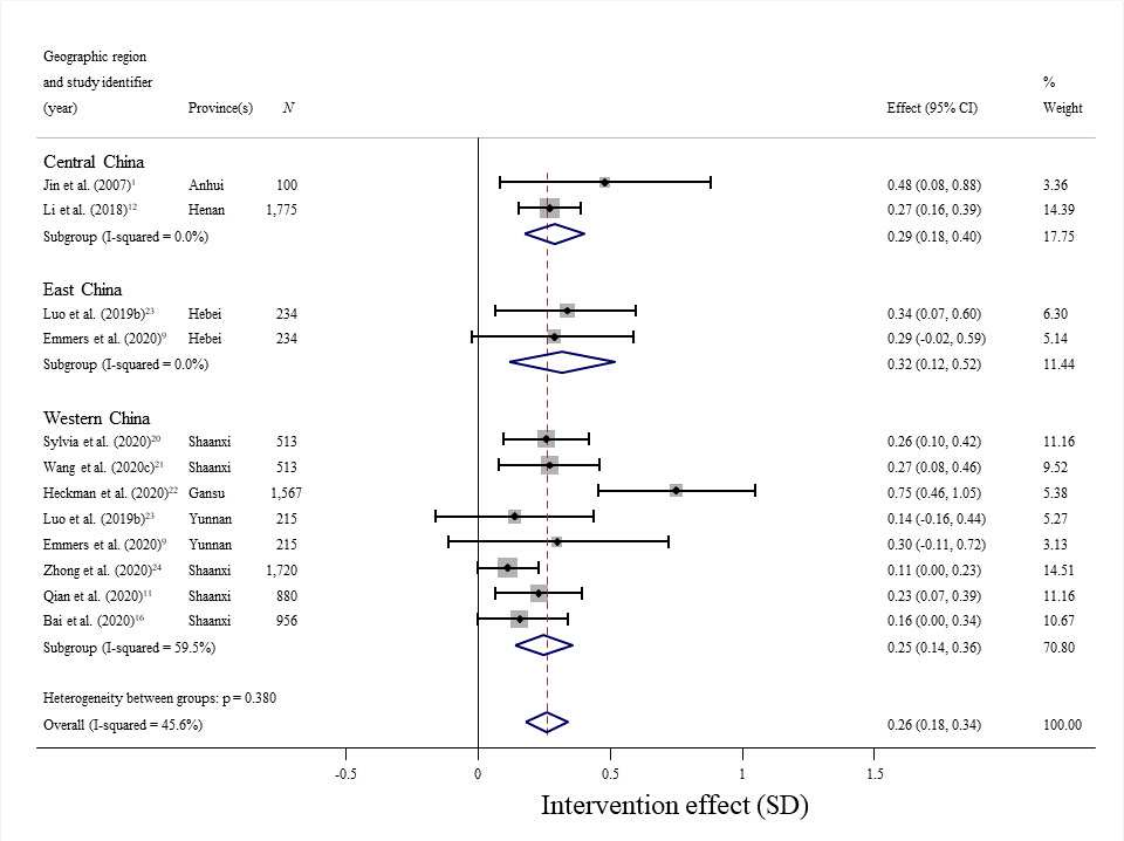
Appendix Figure 6.14: Pooled Prevalence of Early Language Delay by Geographic Region



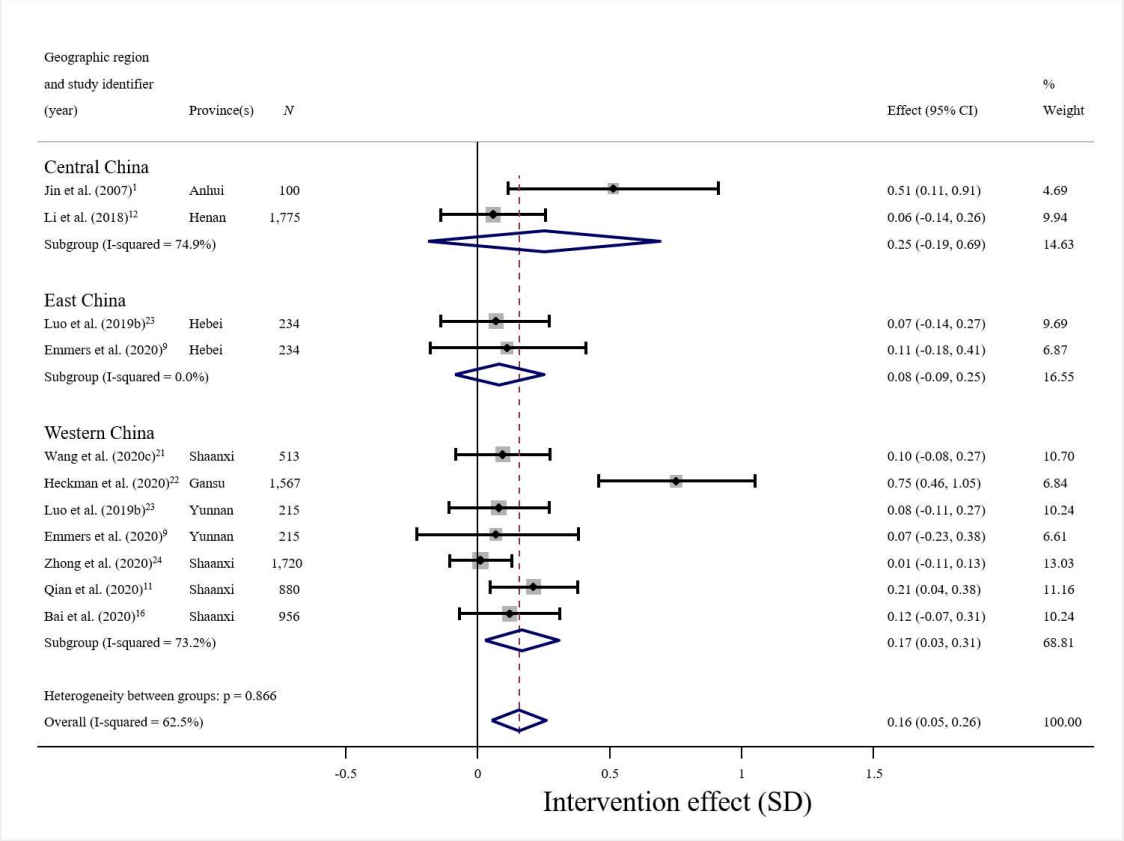
Appendix Figure 6.15: Pooled Prevalence of Early Social-Emotional Delay by Geographic Region



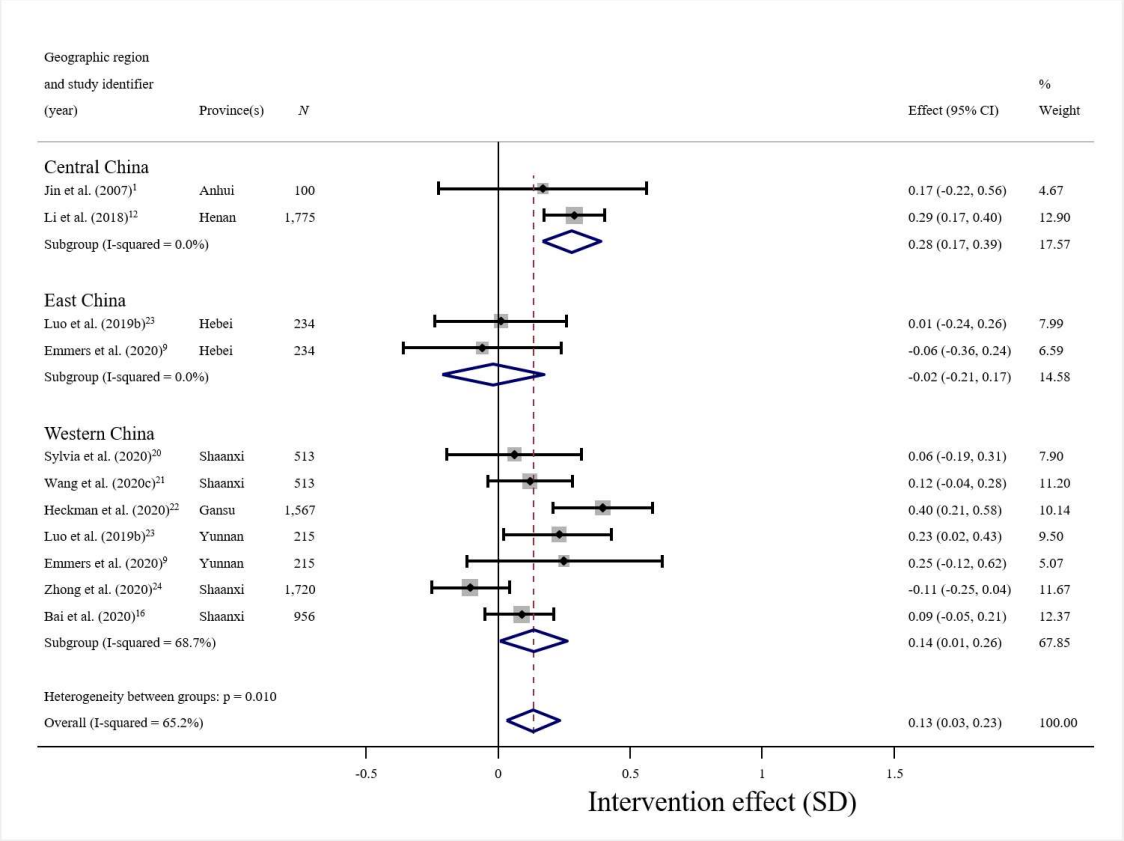
Appendix Figure 6.16: Pooled Intervention Effects on Early Cognitive Development by Geographic Region



Appendix Figure 6.17: Pooled Intervention Effects on Early Language Development by Geographic Region



Appendix Figure 6.18: Pooled Intervention Effects on Early Social-Emotional Development by Geographic Region



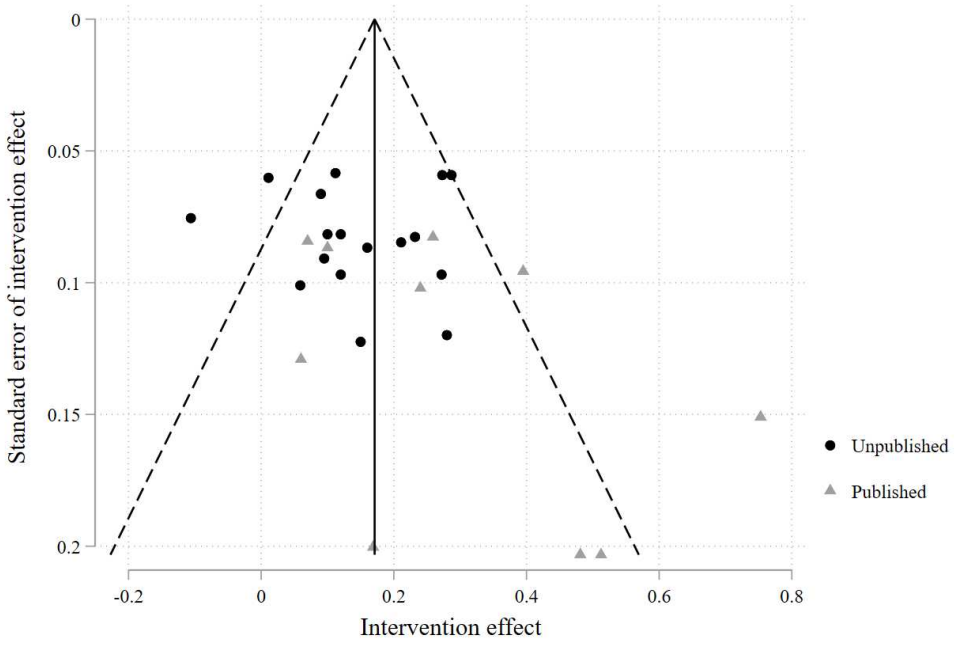
Appendix 7: Unpublished studies

1. Publication bias

We include several unpublished papers and reports in our analyses. Because these studies have not yet undergone peer review, there may be unintentional errors or biases in their data, although it should be noted that we found low levels of bias in all of our formal bias checks. As argued by Dwan et al.[37] empirical research shows that results in published studies are more likely to be positive and statistically significant than are results in unpublished studies. Moreover, publishing studies with insignificant impacts is more difficult and may take longer, which may lead to a "pipeline bias." This type of publication bias is a threat to the validity of systematic reviews.

A common method to test for publication bias is to analyze a funnel plot visually. A funnel plot usually presents effect sizes plotted against their standard errors. Appendix Figure 7.1 shows the funnel plot of the intervention effects in the published and unpublished studies that are included in the meta-analysis of this study. In the presence of publication bias, the funnel plot is expected to be skewed. The funnel plot of the published studies looks more skewed than does the funnel plot of the unpublished studies. The visual examination of asymmetry of funnel plots, however, can be subjective.[38] Therefore, we additionally use the trim-and-fill method and Egger's test to examine the presence of publication bias. First, the trim-and-fill method computes the number of studies that may be missing, and an adjusted intervention effect is derived by "trimming" smaller studies that cause funnel plot asymmetry and imputing studies that may be missing.[39] The trim-and-fill test indicates that no studies are missing from the reviewed literature and that the reported intervention effects are appropriate. Second, Egger's test regresses effect sizes on their precision.[38] In the absence of publication bias, the regression intercept is expected to be zero. It has been shown in the literature that Egger's test is more likely to overestimate the risk of publication bias for studies that assess the impacts on continuous outcome measures.[40] Even though the reviewed studies evaluate impacts on continuous ECD scores, the null hypothesis of the Egger test of no small-study effects is not rejected at the 10% level when we control for between-study heterogeneity induced by differences in ECD measurement tools ($p = 0.12$).

Appendix Figure 7.1: Funnel plot with pseudo 95% confidence limits



2. Actor(s) in charge of program implementation and role of co-authors

We provide information on the actors who were involved in the implementation of the included (published and unpublished) intervention studies and the role of co-authors in these studies.

Appendix Table 7.1: Actor(s) in Charge of Program Implementation and Role of Co-authors

Author(s)	Year	Province(s)	Actor(s) in charge of implementation	Role of co-authors
Jin et al.	2007[1]	Anhui	Research team that consists of members from the Department of Pediatric Development and Behavior of Shanghai Children's Medical Centre, Macfarlane Burnet Institute for Medical Research and Public Health, and School of Medicine of Shanghai Jiaotong University	No involvement
Sylvia et al.	2020[20]	Shaanxi	Family Planning Commission of the People's Republic of China	NW, SS, AM, RL, SR were involved in program evaluation
Wang et al.	2020c[21]	Shaanxi	Family Planning Commission of the People's Republic of China	LW, NW, YQ, SS, SR were involved in program evaluation
Heckman et al.	2020[22]	Gansu	State Council of the People's Republic of China: China Development Research Foundation (CDRF)	No involvement
Luo et al.	2019b[23]	Yunnan, Hebei	Save the Children (Humanitarian Aid Organization)	LR, DE, NW, SS, SR were involved in program evaluation
Emmers et al.	2020[9]	Yunnan, Hebei	Save the Children (Humanitarian Aid Organization)	DE, LR, SS, SR were involved in program evaluation
Zhong et al.	2020[24]	Shaanxi	Local villages and township doctors	RL, SS, SD, AM, SR were involved in program evaluation
Qian et al.	2020[11]	Shaanxi	Local villages and township doctors	QY, JQ, MA, RS were involved in program evaluation
Li et al.	2018[12]	Henan	One Sky (Foundation)	No involvement
Bai et al.	2020[16]	Shaanxi	Hupan Modou Foundation and National Health Commission of the People's Republic of China	DE was involved in program evaluation

Appendix Table 7.2: Summary of Included Studies and their Citations

Panel A. Reviewed studies included in systematic review or meta-analyses

No.	Author(s)	Year of data collection	Province(s)	Citation
1.	Jin et al. (2007)	2003	Anhui	Jin X, Sun Y, Jiang F, et al. "Care for Development" intervention in rural China: a prospective follow-up study. *J Dev Behav Pediatr* 2007;28:213–18.
2.	Ma et al. (2008)	2005–2006	Guangdong	Ma L, Chi L, Su Y, et al. Investigation of development and growth status of 0–18 months children in Guangdong (in Chinese). *Maternal and Child Health Care of China* 2008;13:1844–47.
3.	Yang et al. (2019)	2006–2008	Shaanxi	Yang X, Yin Z, Cheng Y, et al. Features and associated factors of the behavioral development of 24-month-old children in rural China: follow-up evaluation of a randomized controlled trial. *Sci Rep* 2018;8(1):13977.
4.	Yue et al. (2017)	2014	Shaanxi	Yue A, Shi Y, Luo R, et al. China's invisible crisis: cognitive delays among rural toddlers and the absence of modern parenting. *China J* 2017;78:50–80.
5.	Zhou et al. (2019a)	2013	Shanxi, Guizhou	Zhou S, Zhao C, Huang X, et al. The effect of a community-based, integrated and nuturing care intervention on early childhood development in rural China. *Public Health* 2019;167:125–35.
6.	Zhang et al. (2018)	2013	Shanxi, Guizhou	Zhang J, Guo S, Li Y, et al. Factors influencing developmental delay among young children in poor rural China: a latent variable approach. *BMJ Open* 2018;8:e021628.
7.	Wei et al. (2018)	2013	Shanxi, Guizhou	Wei Q, Zhang C, Zhang J, et al. Caregiver's depressive symptoms and young children's socioemotional development delays: a cross-sectional study in poor rural areas of China. *Infant Ment Health J* 2018;39:209–19.
8.	Luo et al. (2019a)	2015	Yunnan, Hebei	Luo R, Jia F, Yue A, et al. Passive parenting and its association with early child development. *Early Child Dev Care* 2019;189:1709–23.
9.	Emmers et al. (2020)	2015	Yunnan, Hebei	Emmers D, Luo R, Rozelle S, et al. Dynamics of early capability formation: evidence from a cluster-randomized parenting experiment in rural China. REAP Working Paper. 2020.
10.	Wang et al. (2020a)	2016	Shaanxi	Wang B, Luo X, Yue A, et al. Family environment in rural China and the link with early childhood development. *Early Child Dev Care* 2020; published online June 29. DOI:10.1080/03004430.2020.1784890.
11.	Qian et al. (2020)	2016	Shaanxi	Qian Y, Jiang Q, Medina A, et al. The impact of group-based early childhood intervention: evidence from below 1-year-old infants in rural China. REAP Working Paper. 2020.
12.	Li et al. (2018)	2016	Henan	Li G, Li Y. An evaluation of an early childhood development intervention in rural communities in Henan province. Henan University Department of Education Working Paper. 2018.
13.	Zhou et al. (2019b)	2016–2017	Jiangxi, Ningxia, Qinghai, Xinjiang	Zhou H, Ding Y, Yang Y, et al. Effects on developmental outcomes after cesarean birth versus vaginal birth in Chinese children aged 1–59 months: a cross-sectional community-based survey. *PeerJ* 2019;7:e7902.
14.	Wang et al. (2019)	2017	Beijing, Henan, Shaanxi	Wang L, Liang W, Zhang S, et al. Are infant/toddler developmental delays a problem across rural China?. *J Comp Econ* 2019;47:458–69.
15.	Zhong et al. (2019)	2017	Guizhou	Zhong J, Gao J, Liu C, Huang J, Luo R. Quantity-quality trade-off and early childhood development in rural family: evidence from China's Guizhou province. *Int J Environ Res Pub He* 2019;16:1307.
16.	Bai et al. (2020)	2018	Shaanxi	Bai Y, Emmers D, Li Y, et al. Effects of a government-led parenting programme on early child development outcomes in rural China: a cluster-randomised controlled trial. Catholic University Working Paper. 2020.
17.	Tan et al. (2020)	2018	Jiangxi, Guizhou, Sichuan, Hebei, Henan	Tan C, Zhao C, Dou Y, et al. Caregivers' depressive symptoms and social–emotional development of left-behind children under 3 years old in poor rural China: the mediating role of home environment. *Child Youth Serv Rev* 2020;116:105109.
18.	Wang et al. (2020b)	2019	Yunnan	Wang L. Rural early childhood development delays and parenting practices in yunnan province: data from baseline study sponsored by Save-the-Children. Shaanxi Normal University Working Paper. 2020.

19.	Emmers & Wang (2020)	2019	Shaanxi	Emmers D, Wang L, Sylvia S, et al. Rural-urban differences in the intergenerational transmission of cognitive capabilities in China. REAP Working Paper. 2020.
20.	Sylvia et al. (2020)	2014	Shaanxi	Sylvia S, Warrinnier N, Luo R, et al. From quantity to quality: delivering a home-based parenting intervention through China's family planning cadres. *Econ J* 2020; published online September 14. DOI:10.1093/ej/ueaa114.
21.	Wang et al. (2020c)	2014	Shaanxi	Wang L, Warrinnier N, Qian Y, et al. Persistence and fade-out: the impact of a short-term early childhood intervention at school entry. REAP Working Paper. 2020.
22.	Heckman et al. (2020)	2015	Gansu	Heckman JJ, Liu B, Lu M, et al. Treatment effects and the measurement of skills in a prototypical home visiting program. National Bureau of Economics Research (NBER) Working Paper Series. 2020. https://www.nber.org/papers/w27356 (accessed November 15, 2020).
23.	Luo et al. (2019b)	2015	Yunnan, Hebei	Luo R, Emmers D, Warrinnier N, et al. Using community health workers to deliver a scalable integrated parenting program in rural China: a cluster-randomized controlled trial. *Soc Sci Med* 2019;239:112545.
24.	Zhong et al. (2020)	2016	Shaanxi	Zhong J, Luo R, Sylvia S, et al. Passive versus active service delivery: comparing the effects of two parenting interventions on early cognitive development in rural China. REAP Working Paper. 2020.

Panel B. Other references

25. Huang H, Tao S, Zhang Y, et al. Standardization of Bayley Scales of Infant Development in Shanghai. *Chin J Child Health* 1993;1(3):158–160.
26. Yi S, Luo X, Yang Z, et al. The revising of Bayley Scales of Infant Development (BSID) in China. *Chin J Clin Psychol* 1993;1:71–75.
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